

Supporting Information

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I. SUPPLEMENTARY METHODS

A. Computational details

Supplementary Table I: Computational parameters used for each material: basis set energy cut-off, **k**-point grid for bulk (b) structure and supercell (s) defect calculations, supercell expansions of either the primitive (p) or the conventional (c) unit cells, percentage of exact HF exchange (α) used within the HSE hybrid DFT functional and the valence electrons considered for each element. The scheme of Grimme et al.¹ (D3) was included to account for the van der Waals interactions present in Sb₂S/Se₃.

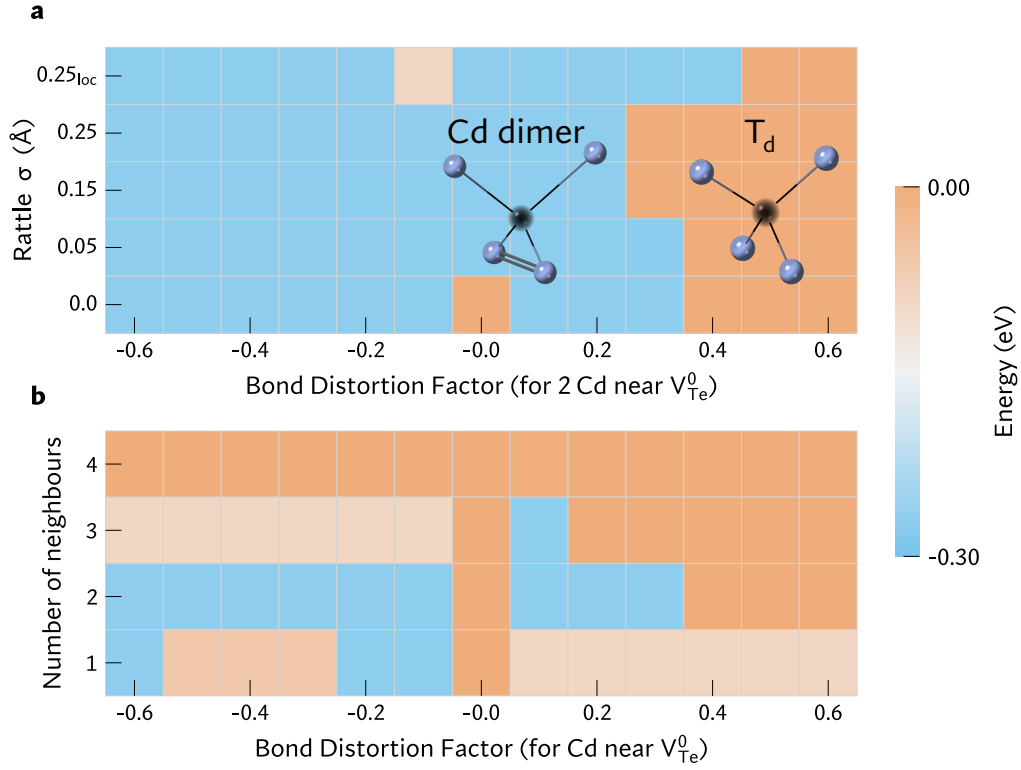
Material	Cut-off (eV)	k-grid	Supercell	α %	Valence electrons
CdTe	450	4,4,4 (b,c) 2,2,2 (s)	2,2,2 (c)	0.345	Cd: 4d ¹⁰ 5s ² Te: 5s ² 5p ⁴
GaAs	300	7,7,7 (b,p) 2,2,2 (s)	3,3,3 (p)	0.280	Ga: 3d ¹⁰ 4s ² 4p ¹ As: 3s ² 3p ³ Si: 3s ² 3p ² Sn: 5s ² 4d ¹⁰ 5p ² S: 3s ² 3p ⁴ Te: 5s ² 5p ⁴
CeO ₂	550	4,4,4 (b,c) 2,2,2 (s)	2,2,2 (c)	0.250	Ce: 5s ² 5p ⁶ 4f ¹ 5d ¹ 6s ² O: 2s ² 2p ⁴ Be: 2s ² Ni: 3d ⁹ 4s ¹ Pd: 4d ¹⁰ Pt: 5d ⁹ 6s ¹ Cu: 3d ¹⁰ 4s ¹
Sb ₂ Se ₃	300	4,2,2 (b,c) 2,2,2 (s)	3,1,1 (c)	0.250 + D3	Sb: 5s ² 5p ³ Se: 4s ² 4p ⁴
Sb ₂ S ₃	300	3,2,2 (b,c) 1,2,2 (s)	3,1,1 (c)	0.250 + D3	Sb: 5s ² 5p ³ S: 3s ² 3p ⁴
In ₂ O ₃	500	3,3,3 (b,c) 3,3,3 (s)	1,1,1 (c)	0.250	In: 4d ¹⁰ 5s ² 5p ¹ O: 2s ² 2p ⁴
ZnO	500	12,12,6 (b,c) 4,4,3 (s)	3,3,2 (c)	0.250	Zn: 3d ¹⁰ 4s ² O: 2s ² 2p ⁴
Si	300	4,4,4 (b,c) 2,2,2 (s)	2,2,2 (c)	0.110	Si: 3s ² 3p ²
a-TiO ₂	500	4,4,2 (b,c) 2,2,2 (s)	3,3,1 (c)	0.250	Ti: 3p ⁶ 3d ² 4s ² O: 2s ² 2p ⁴
r-TiO ₂	500	4,3,3 (b,c) 2,2,2 (s)	4,2,2 (c)	0.250	Ti: 3p ⁶ 3d ² 4s ² O: 2s ² 2p ⁴

B. Bulk crystals

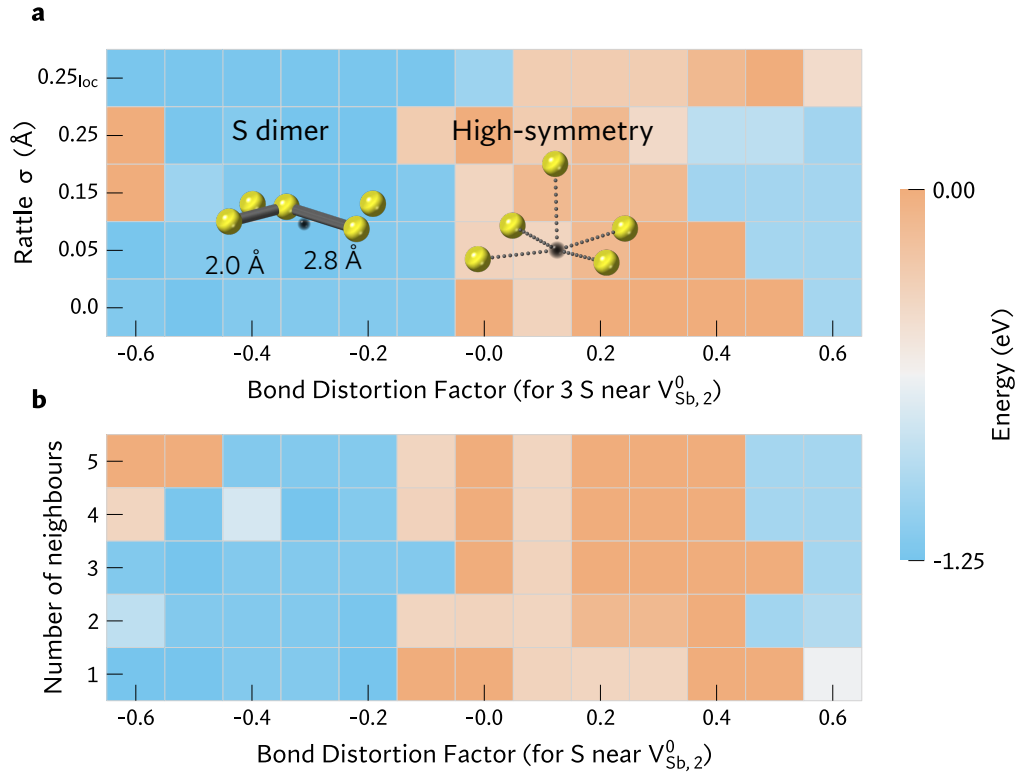
Supplementary Table II: Calculated lattice parameters (in Å) of the conventional cells for the studied materials and percentage differences from the experimental lattice parameters.

Material	a	$\Delta a\%$	b	$\Delta b\%$	c	$\Delta c\%$	Ref.
CdTe	6.543	1	6.543	1	6.543	1	Strauuss ²
GaAs	5.662	-0.1	5.662	-0.1	5.662	-0.1	Madelung ³
CeO ₂	5.394	-0.3	5.394	-0.3	5.394	-0.3	Rossignol ⁴
Sb ₂ Se ₃	3.961	-0.6	11.494	-1.3	11.928	1.1	Voutsas ⁵
Sb ₂ S ₃	3.841	0.1	11.063	-1.5	11.362	0.4	Bayliss ⁶
In ₂ O ₃	10.205	0.8	10.205	0.8	10.205	0.8	González ⁷
ZnO	3.25	-0.04	3.25	-0.04	5.20	-0.18	Kisi ⁸
Si	5.449	0.3	5.449	0.3	5.449	0.3	Bond ⁹
a-TiO ₂	3.774	-0.3	3.774	-0.3	9.579	0.7	Burdett ¹⁰
r-TiO ₂	2.950	-0.3	4.597	0.05	4.597	0.05	Shiiba ¹¹

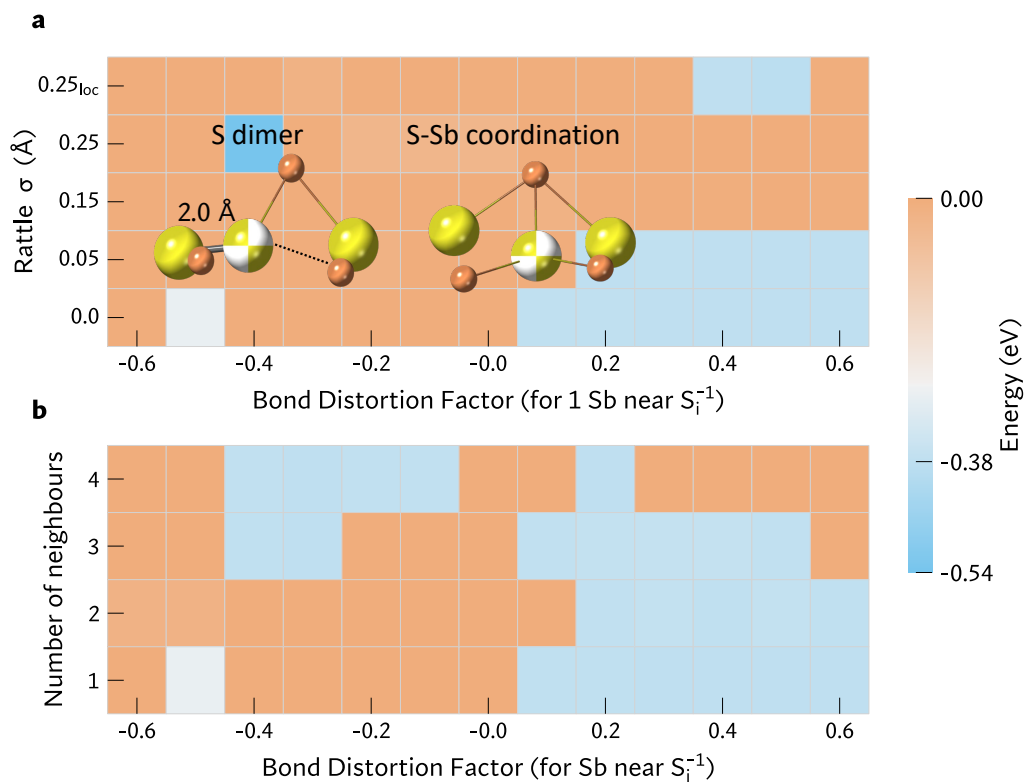
C. Defect Structure Searching (ShakeNBreak) approach: Parameter optimisation



Supplementary Figure 1: Relative energy of final structures for V_{Te}^0 in CdTe (either ‘Cd dimer’ (-0.3 eV), ‘Tetrahedral’ (0 eV) or a C_{3v} reconstruction (-0.1 eV)), for different rattle standard deviations (**a**) and number of distorted defect neighbours (**b**). The wider interval of bond distortions leading to the defect ground state is obtained with a standard deviation of 0.25 Å localised to the atoms within 5 Å from the defect (0.25_{loc}). Regarding the number of neighbours, best performance is obtained when distorting two cadmium atoms, as expected considering that the defect has two extra electrons in the Cd dangling bonds.

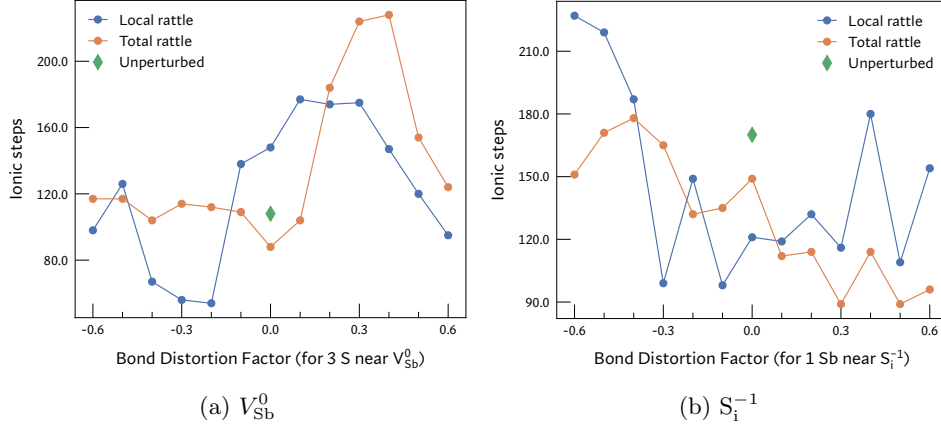


Supplementary Figure 2: Relative energy of final structures for $V_{\text{Sb},2}^0$ in Sb_2S_3 (either 'S dimer' (-1.25 eV) or 'High-symmetry' configuration (0 eV)), for different rattle standard deviations (**a**) and number of distorted defect neighbours (**b**). Here, the bond distortion parameters have a minor influence (with two and three distorted neighbours performing slightly better), which suggests that any symmetry breaking is enough to escape the local minimum and find the favourable 'S dimer' configuration.



Supplementary Figure 3: Relative energy of final structures for S_i^{-1} in Sb_2S_3 (either two different arrangements of ‘S dimer’ (-0.55 eV, -0.38 eV) or weak S-Sb coordination (0 eV)), for different rattle standard deviations (**a**) and number of distorted defect neighbours (**b**). Notably, the lowest energy structure is only obtained when applying a total rattle standard deviation of 0.25 $\text{\AA}$. In this case, the number of distorted neighbours seems to have a minor role – with three and one performing slightly better.

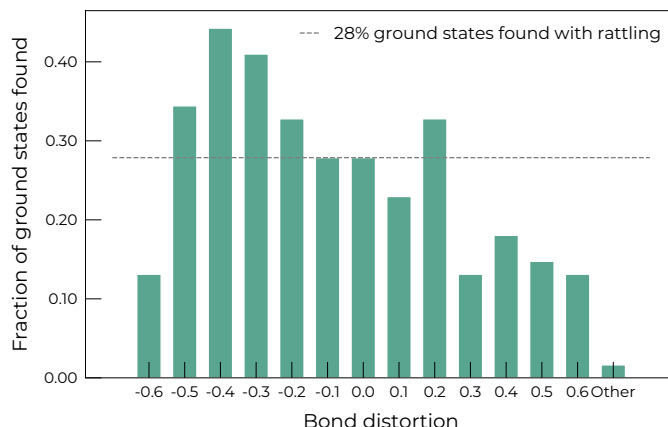
1. Comparison of local and total rattle



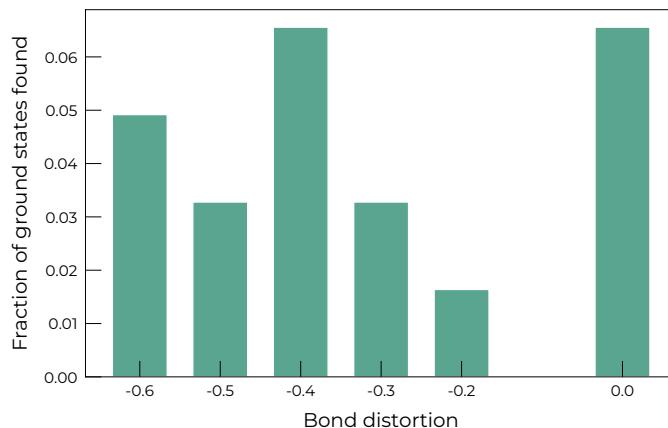
Supplementary Figure 4: Total number of ionic steps for each bond distortion with a local (restricted to 5 Å sphere about the defect, $\sigma = 0.25$) and a total rattle ($\sigma = 0.25$) in the relaxation of V_{Sb}^0 (a) and S_i^{-1} (b) in Sb_2S_3 . Surprisingly, the local rattle does not lead to a significant decrease in the number of ionic steps compared to its total counterpart.

2. Comparison of bond distortions

To investigate if a smaller subset of distortions could identify most energy lowering reconstructions, we compared the number of ground states found by each distortion for all systems studied (total of 61 energy lowering reconstructions) (Fig. 5). Further, we also investigated the cases where only one distortion finds the ground state (Fig. 6). Based on this, for cases where computational limitations require a sparser grid, we recommend using the following bond distortions (-60%, -40%, -30%, -20%, 0%, 20%, 40%), or a subset from these by considering Fig. 5. We note that only using a rattling distortion (random perturbations to all atoms in the cell) misses more than 70% of the identified energy lowering distortions (dashed line in Fig. 5), which emphasises the importance of using bond distortions.



Supplementary Figure 5: Fraction of ground state structures identified by each distortion for a total of 61 energy lowering reconstructions. The label “Other” corresponds to using the ground state structure identified for one charge state as a candidate configuration for other charge states of the same defect (where that configuration was not found). The dashed grey line shows the fraction of ground states identified when only applying random distortions to all atoms in the cell (Bond distortion=0.0).



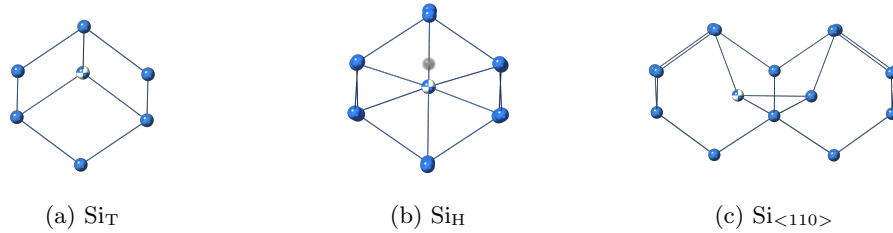
Supplementary Figure 6: Set of distortions which uniquely find the ground state (e.g. the only ones from their distortion grid in finding the lowest energy structure), and fraction of ground states identified by each of them (out of a total of 61 energy lowering reconstructions).

D. Comparison with other structure searching approaches

To compare the performance of our method with existing approaches in the literature, it was applied to the neutral silicon self-interstitial and the neutral oxygen vacancy in anatase. These systems were selected as they were previously investigated using the evolutionary approach of Arrigoni and Madsen.¹²

1. Silicon self-interstitial

The high technological relevance of the silicon self-interstitial has motivated a large number of studies on its structural and migration properties. These have identified several low energy configurations: Si in an tetrahedral interstitial (Si_T), Si in a hexagonal interstitial (Si_H) and the split $\langle 110 \rangle$ configuration ($\text{Si}_{\langle 110 \rangle}$), where two Si move from their original site and form a dumbbell configuration along the $\langle 110 \rangle$ direction^{13–24}. These are shown in Fig. 7. Previous studies agree that the hexagonal and split configurations are the most stable, differing by a small energy difference (order of meV), with the latter likely corresponding with the ground state^{13,21,22,25–27}. Significantly higher in energy lies the tetrahedral structure (≈ 0.4 eV)^{18–20,25}.



Supplementary Figure 7: Lowest energy structures for the silicon self-interstitial: a) Si_T b) Si_H and c) $\text{Si}_{\langle 110 \rangle}$. The silicon interstitial is shown with a different pattern and the original tetrahedral position is shown in grey.

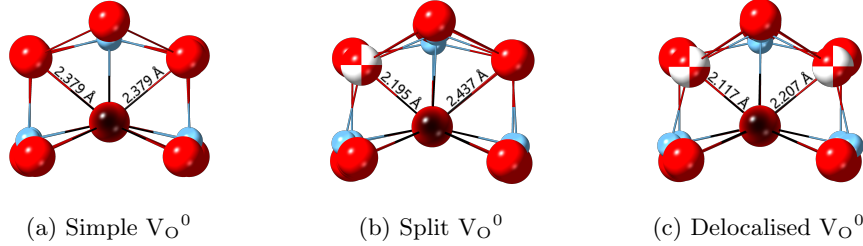
By applying our method, we identified the two lowest energy structures, thereby exhibiting similar results to the evolutionary approach of Arrigoni and Madsen, yet with a simpler implementation. In agreement with their result, we identified the ground state as the hexagonal configuration. We find it to lie 7 meV lower in energy than the split arrangement, opposed to their result of 23 meV using local DFT (LDA) and Γ -point only $\mathbf{k}$ -point sampling. As expected, the latter agrees better with our initial result (31 meV), obtained prior to optimising the final structures with a denser $\mathbf{k}$ -mesh. On this point, it is noteworthy how the approximate exploration of the PES (i.e. the Γ -point approximation) qualitatively described the landscape despite the minute energy differences between local minima. These minor energy differences between structures explain the debate across different studies (and energy functionals) regarding the actual ground state.^{18,23–25,28} Semi-local functionals often favour the split configuration^{13,22,23,25–27}, while higher levels of theory (QMC²⁴ or G_0W_0 corrections¹⁸) agree that the hexagonal structure lies lower in energy, by 160 and 60 meV, respectively.

Finally, we note that a tetrahedral configuration (Si_T) was found to have no local stability with our hybrid DFT functional, with unperturbed relaxation of this initial geometry without symmetry constraints yielding the hexagonal arrangement (Si_H), as also reported in other studies²⁴.

2. Neutral oxygen vacancy in anatase

The key role of anatase in photo-catalysis^{29,30}, solar cells^{31,32}, spintronic devices³³ and as a promising transparent conductive oxide^{34,35} have motivated intense study of its intrinsic defects³⁶. In particular, significant efforts have been dedicated to the oxygen vacancy^{37–47}, as it renders intrinsic anatase n-type⁴⁸. Several studies^{41–43,46} have reported two possible configurations: the simple vacancy (analogous to the ideal defect structure, Fig. 8 (a)) and the split configuration, where one of the neighbouring oxygen atoms moves towards the vacancy (Fig. 8 (b)); with the latter lying 0.38 eV lower in energy (using HSE(15%))⁴⁶. In terms of the electronic structure, the former is characterised by two localised electrons in the vacancy site while for the latter, one electron is localised on a neighbouring Ti while

the other one is excited to a delocalised conduction band state. Notably, this lower energy configuration has been missed by previous hybrid studies^{45,49}, as a standard relaxation from the ideal structure gets trapped on the simple vacancy basin.

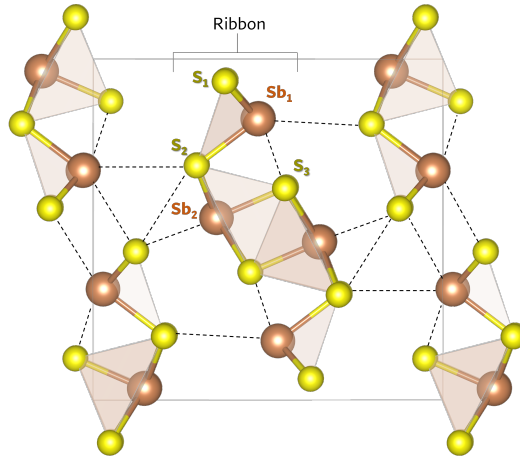


Supplementary Figure 8: Lowest energy structures for the neutral oxygen vacancy in anatase TiO_2 : a) Simple VO^0 b) Split VO^0 and c) Delocalised VO^0 . Ti atoms in blue, O in red, VO in black and displaced O neighbours represented with a different pattern.

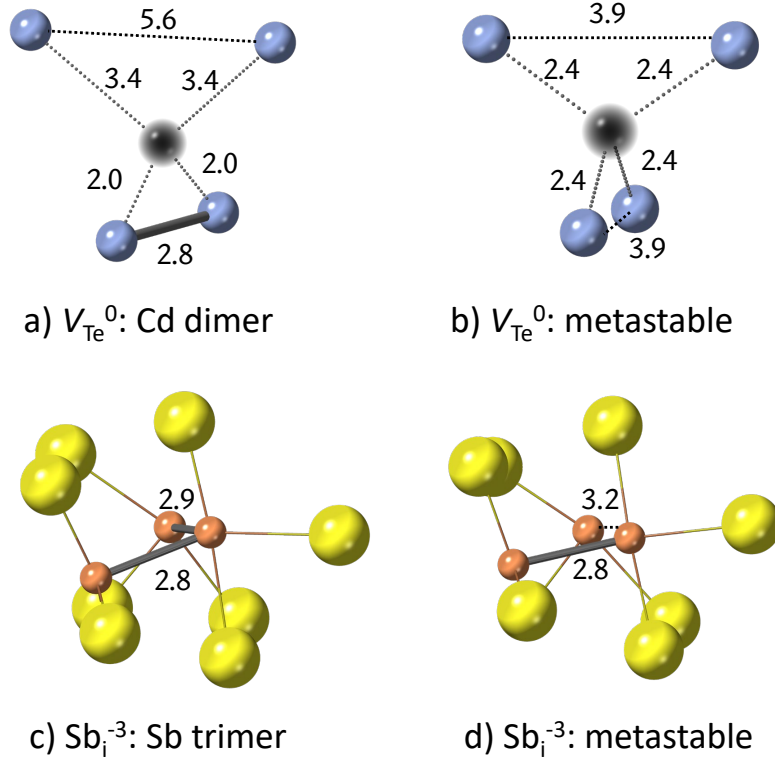
By applying their evolutionary algorithm, Arrigoni and Madsen identified, besides these structures, a new metastable configuration, termed the delocalised vacancy since the two electrons are delocalised in the conduction band (Fig. 8 (c)). Lying 59 meV higher in energy than the split structure, it is characterised by two neighbouring oxygen atoms moving towards the vacancy. These reduce their bond lengths from an initial value of 2.45 Å to 2.12 and 2.21 Å. In comparison, our method also identified these structures, finding the simple vacancy and delocalised vacancy to lie 0.37 and 0.03 eV higher in energy than the split structure, respectively – in good agreement with the result of Arrigoni (0.38, 0.06 eV). Hence, both approaches rendered similar configurations and relative stabilities.

II. SUPPLEMENTARY NOTES

A. Symmetry breaking reconstructions



Supplementary Figure 9: The crystal structure of Sb_2S_3 (and Sb_2Se_3) comprises chains of $[\text{Sb}_4\text{S}_6]_n$ ($[\text{Sb}_4\text{Se}_6]_n$) (termed ribbons), which are bonded by weak van der Waals forces (dashed lines). Each ribbon has two and three symmetry inequivalent sites for antimony and sulfur, respectively, with the Sb cations occupying the centre of a pyramid formed by the closest S anions.

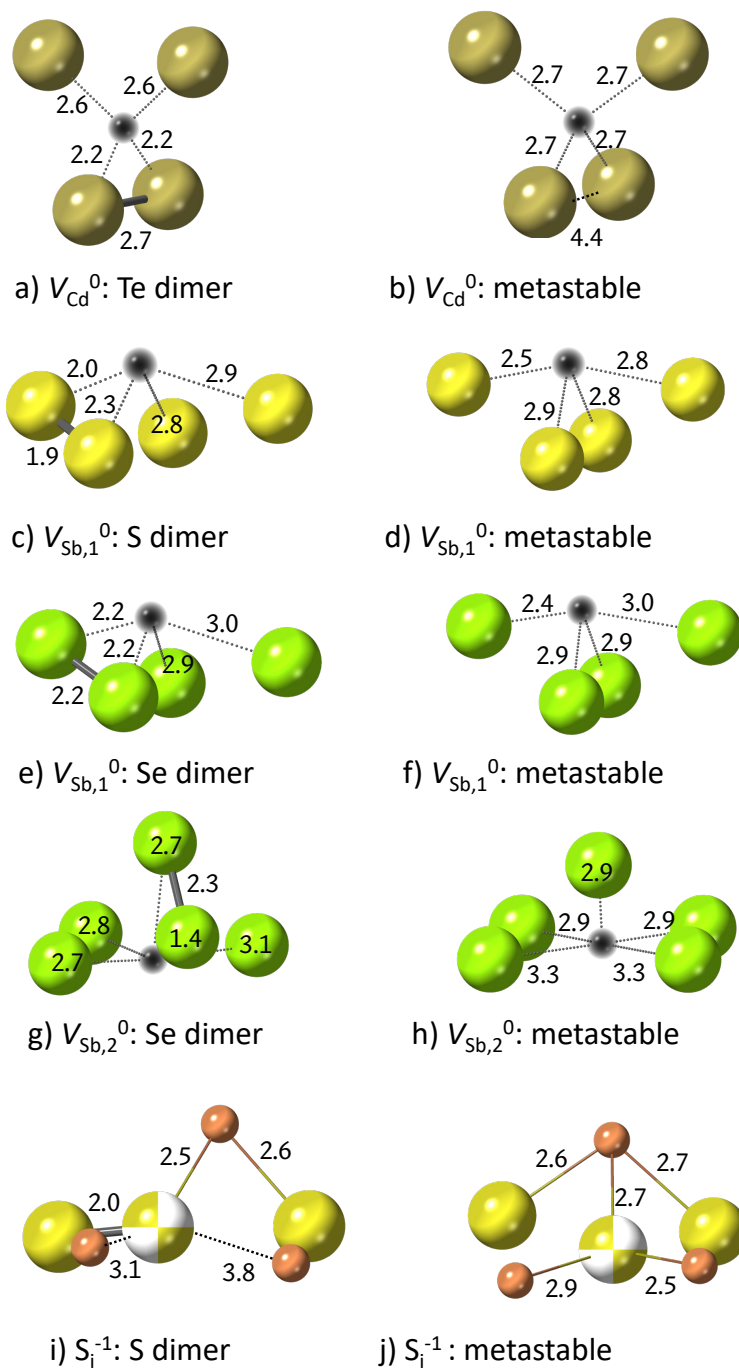


Supplementary Figure 10: Ground state (left: a,c) and metastable (right: b,d) structures identified by our method and by relaxing the high-symmetry configuration, respectively, for cation-cation dimerisation. (Energies and displacements reported in Table 1). Vacancy in black, Cd in blue, Sb in orange and S in yellow.

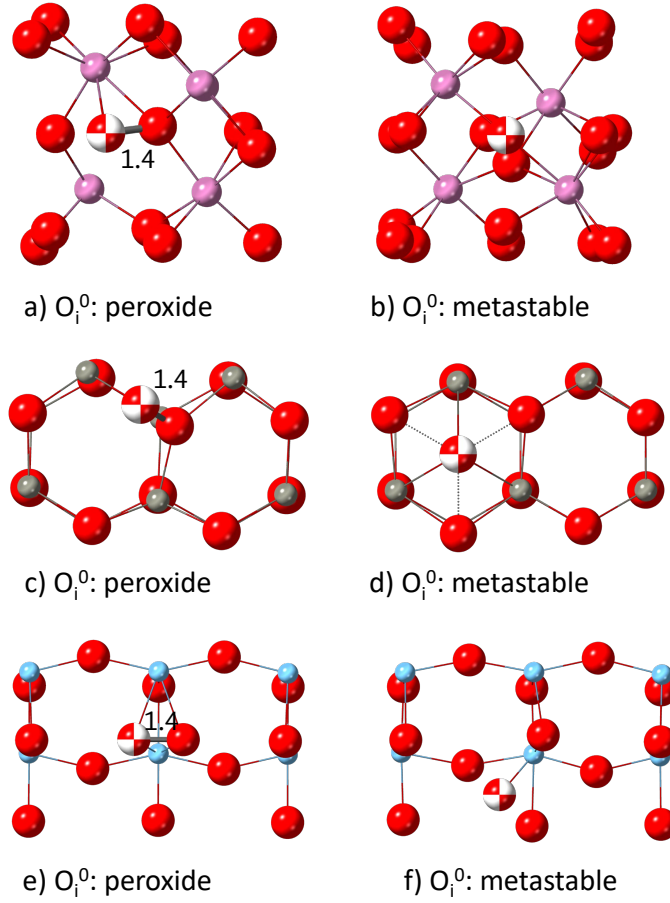
B. Rebonding: dimerisation

Supplementary Table III: Integrated Crystal Orbital Hamilton Population (ICOHP) for the S-S/Se-Se bonds formed by the under-coordinated neighbours of V_{Sb} in $\text{Sb}_2\text{S}_3/\text{Se}_3$.

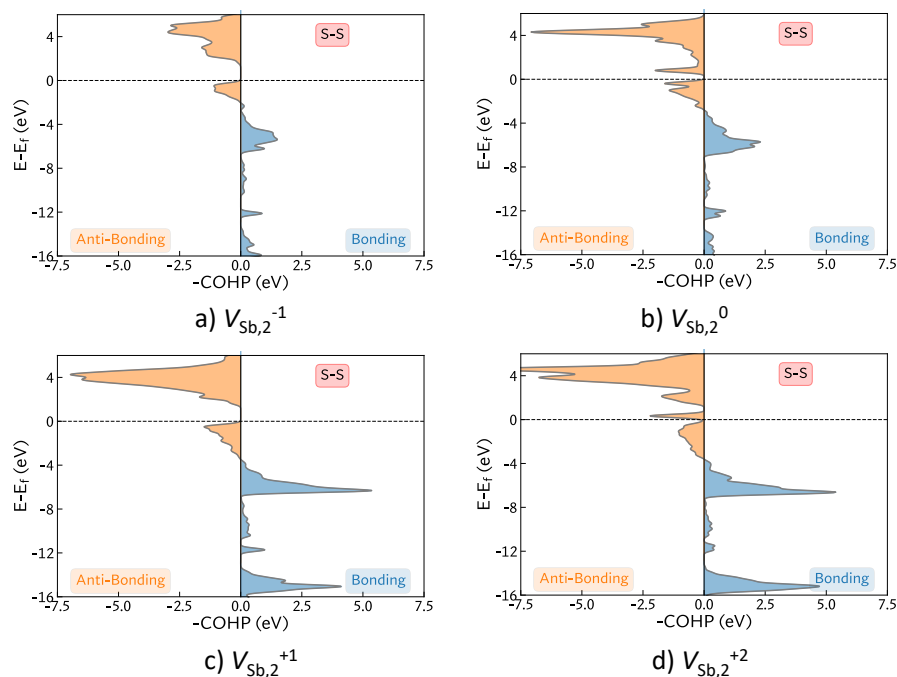
Charge	ICOHP (eV)			
	Sb_2S_3		Sb_2Se_3	
	Site 1	Site 2	Site 1	Site 2
-1	-6.87	-6.84	-4.97	-4.96
0	-9.15	-7.74	-6.60	-5.54
+1	-13.36	-13.41	-6.69	-10.12
+2	-14.50	-14.84	-11.09	-10.83



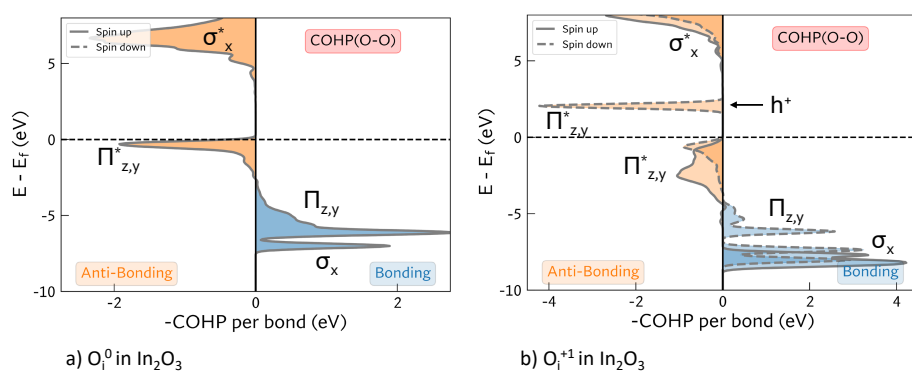
Supplementary Figure 11: Ground state (left: a,c,e,g,i) and metastable (right: b,d,f,h,j) structures identified by our method and by relaxing the high-symmetry configuration, respectively, for anion-anion dimerisation. (Energies and displacements reported in Table 1). Te in gold, Sb in orange, S in yellow and Se in green. Vacancy shown in shaded black and S_i displayed in a different pattern. Pseudo-bonds between a vacancy and its neighbours shown with dotted lines, and distances in Å.



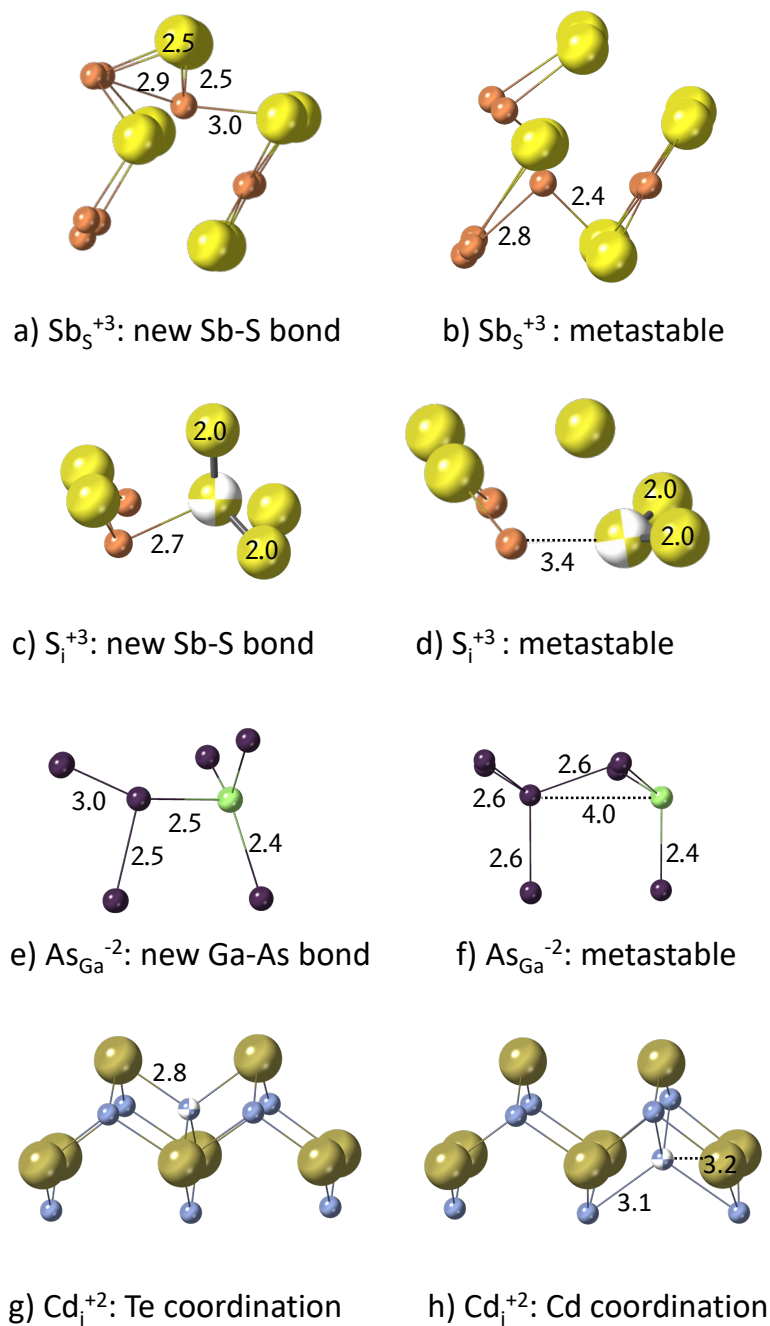
Supplementary Figure 12: Ground state (left: a,c,e) and metastable (right: b,d,f) structures identified by our method and by relaxing the high-symmetry configuration, respectively, for the anion-anion dimerisation in In_2O_3 , ZnO and anatase TiO_2 . (Energies and displacements reported in Table 1). In in purple, Zn in grey, Ti in blue and O in red with O_i shown in a different pattern for clarity.



Supplementary Figure 13: Crystal Orbital Hamilton Population (COHP) analysis for the sulfur-sulfur bonds formed by the vacancy neighbours in the ground state structures of $V_{\text{Sb},2}$ in Sb_2S_3 , illustrating the stronger bonding character of the S-S dimers for the positive charge states.

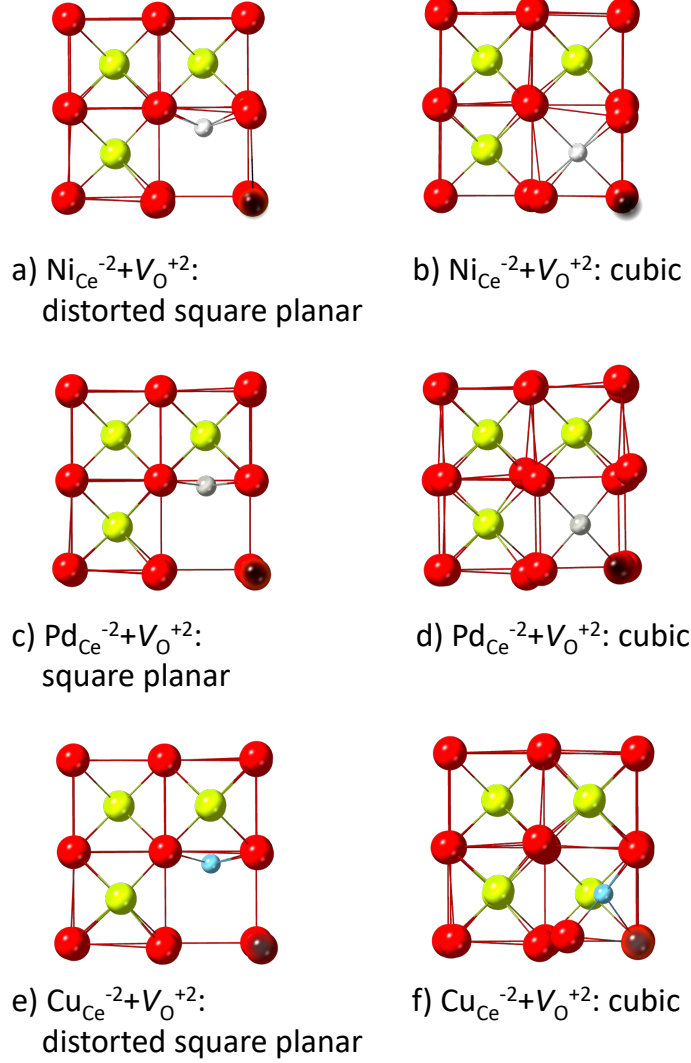


Supplementary Figure 14: Crystal Orbital Hamilton Population (COHP) analysis for the oxygen dimer bonds in the ground state structures of O_i in In_2O_3 . The orbital character of the different states is assigned by comparison with the molecular orbital diagram of a peroxide anion. For positive charge state, the hole is localised on the antibonding π^* state, which appears within the band gap.

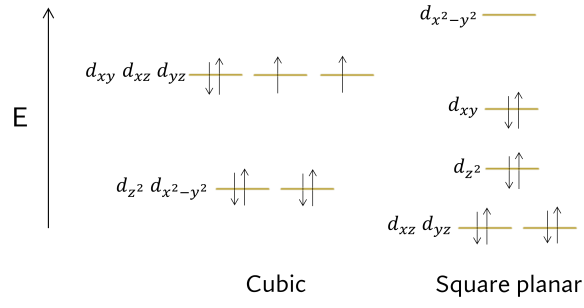


Supplementary Figure 15: Ground state (left: a,c,e,g) and metastable (right: b,d,f,h) structures identified by our method and by relaxing the high-symmetry configuration, respectively, for the cation-anion rebonding reconstruction. (Energies and displacements reported in Table 2). Vacancy in black, Te in gold, Cd in blue, Sb in orange, S in yellow, Ga in green and As in purple. Interstitials shown with a different pattern and distances in Å.

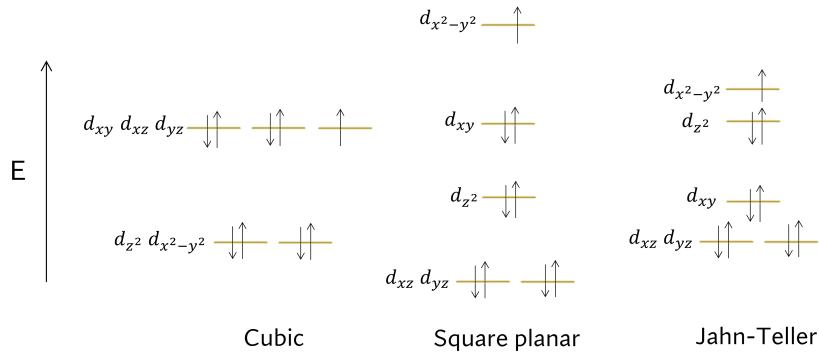
C. Crystal field and Jahn-Teller effects



Supplementary Figure 16: Ground state (left: a,c,e) and metastable (right: b,d,f) structures identified by our method and by relaxing the high-symmetry configuration, respectively, for crystal field and Jahn-Teller reconstructions in ceria. (Energies and displacements reported in Table 3). V_{O} in black, Ce in green, O in red, Ni in light grey, Pd in dark grey and Cu in blue.

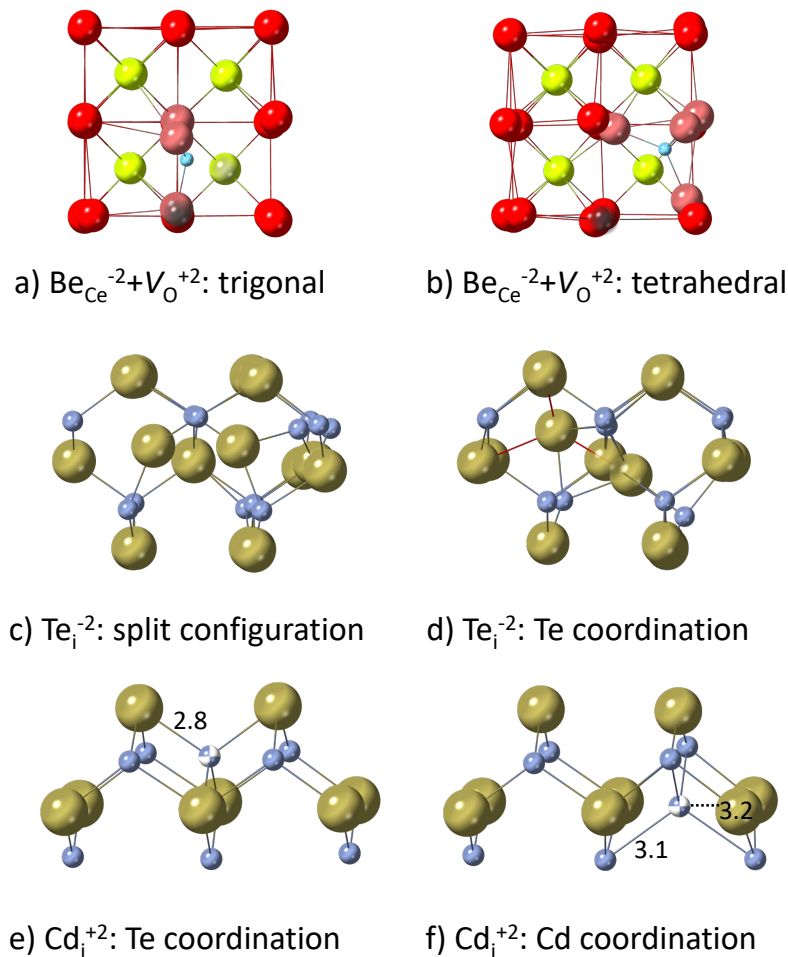


Supplementary Figure 17: Electron energy level diagram for d^8 dopants on CeO_2 showing the crystal field splitting for the initial cubic coordination and the more favourable square-planar arrangement, which leads to a gain in electronic energy.



Supplementary Figure 18: Electron energy level diagram for a d^9 dopant, showing the advantage of the Jahn-Teller distortion as it lowers the energy of the partially-occupied $d_{x^2-y^2}$ orbital, in comparison with the square-planar arrangement.

D. Electrostatically driven



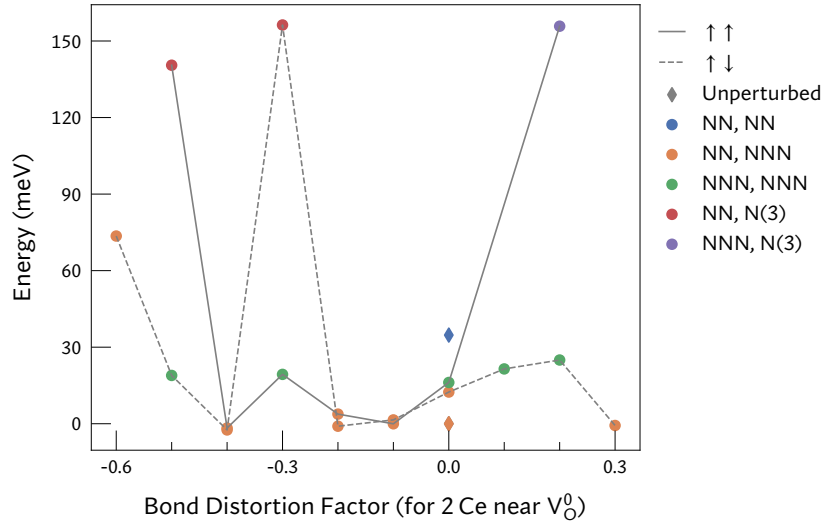
Supplementary Figure 19: Ground state (left: a,c,e) and metastable (right: b,d,f) structures identified by our method and by relaxing the high-symmetry configuration, respectively, for electrostatically driven reconstructions. (Energies and displacements reported in Table 4). Be in blue, Ce in green, O in red, Te in gold and Cd in blue. Vacancy shown in shaded black.

E. Defect bound polarons

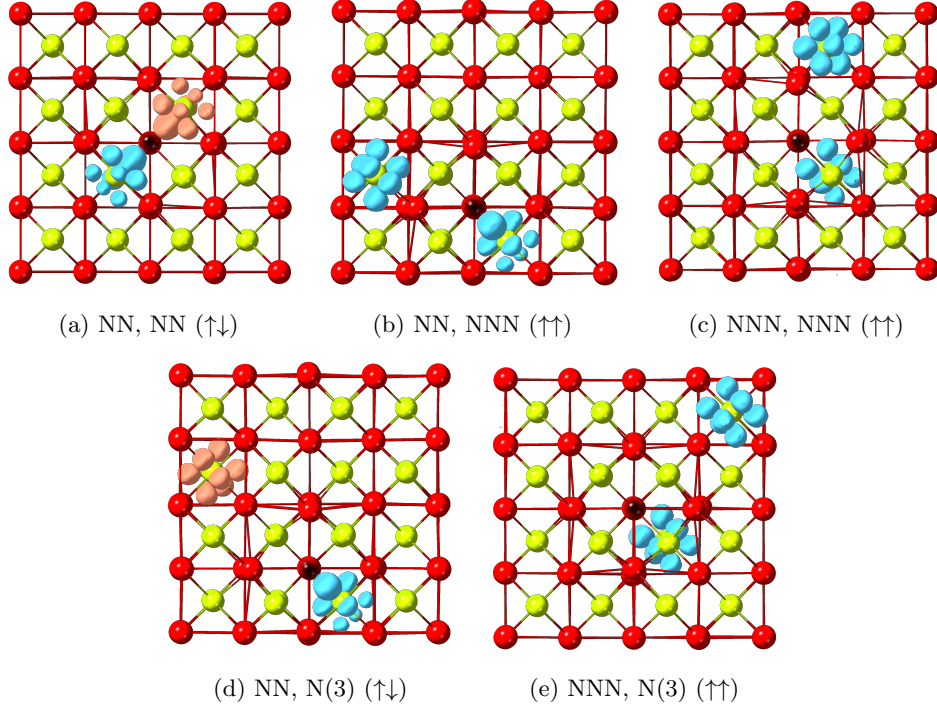
1. Neutral oxygen vacancy in CeO_2

In order to test the ability of the method to identify defect bound polarons, we first considered the neutral oxygen vacancy in CeO_2 . Upon vacancy formation, two electrons localise on two Ce ions, reducing them to Ce(III) . The location of the Ce(III) ions is however not clear, with some studies reporting both Ce(III) as nearest neighbours of the vacancy

(NN, NN)^{50–53}, next nearest neighbours (NNN, NNN)^{54,55}, or even a combination of the two (NN, NNN)⁵⁶. We therefore applied the method to this defect to test whether it was able to identify these configurations. In order to thoroughly sample both spin alignments, we applied it twice: setting the number of unpaired electrons to 2 (spin-aligned, $\uparrow\uparrow$) and also to 0 (anti-aligned, $\uparrow\downarrow$) (Fig. 20, 21). This results in all previously reported configurations being identified: both Ce(III) in the first coordination shell (NN, NN), both in the second coordination shell (NNN, NNN), one in the first and the other in the second (NN, NN) and also combinations of first/second and third coordination shells (NN, N(3) and NNN, N(3)). The energy differences between these states are of the order of meV, explaining the debate in the literature regarding the most stable configuration. These results demonstrate the ability of the method to identify different polaronic states, even when their energy differences are small. In such cases, where the energy differences are within the error of the Γ point approximation, each low energy structures obtained from the initial search should be optimised with denser reciprocal space sampling.



Supplementary Figure 20: Relative energy (in meV) of the different configurations identified for the neutral oxygen vacancy in CeO₂. The method was applied both constraining the difference of the number of electrons in the up and down spin component to 2 (solid line) and also to 0 (dashed line). Standard relaxation from the high symmetry structure shown with a diamond (Unperturbed). The different colours indicate where the two Ce(III) ions are located relative to the vacancy: nearest neighbour (NN), next nearest neighbour (NNN) or in the third coordination shell (N(3)).

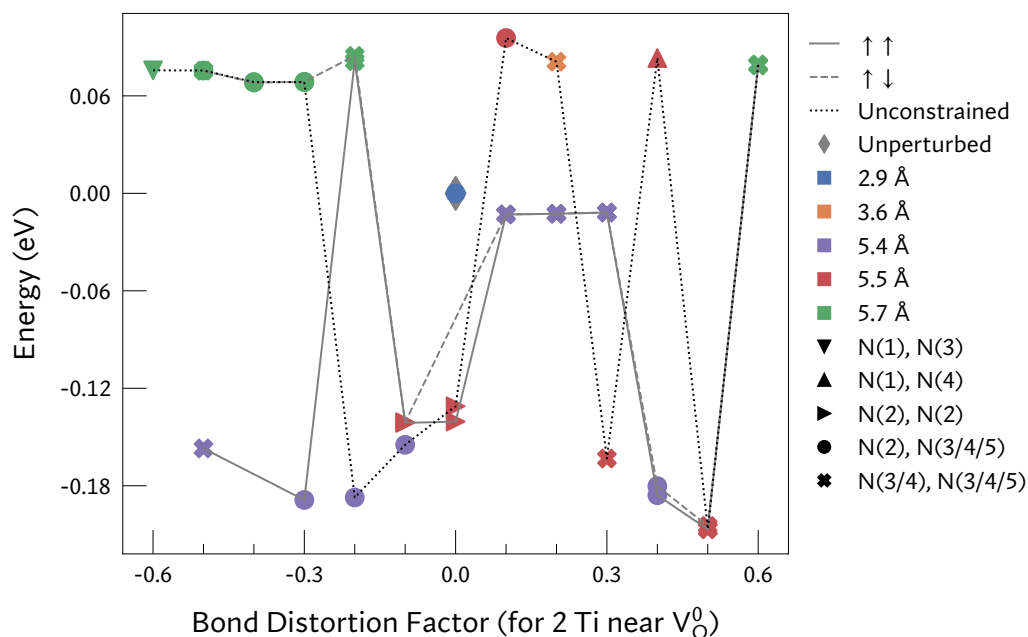


Supplementary Figure 21: Structure with spin density plot for the different configurations identified with our method for V_O^0 in CeO_2 . V_O shown in black, Ce in green and O in red. The spin density of different channels is shown with different colours. In all cases, the isosurfaces correspond to 10.5% of the maximum charge density.

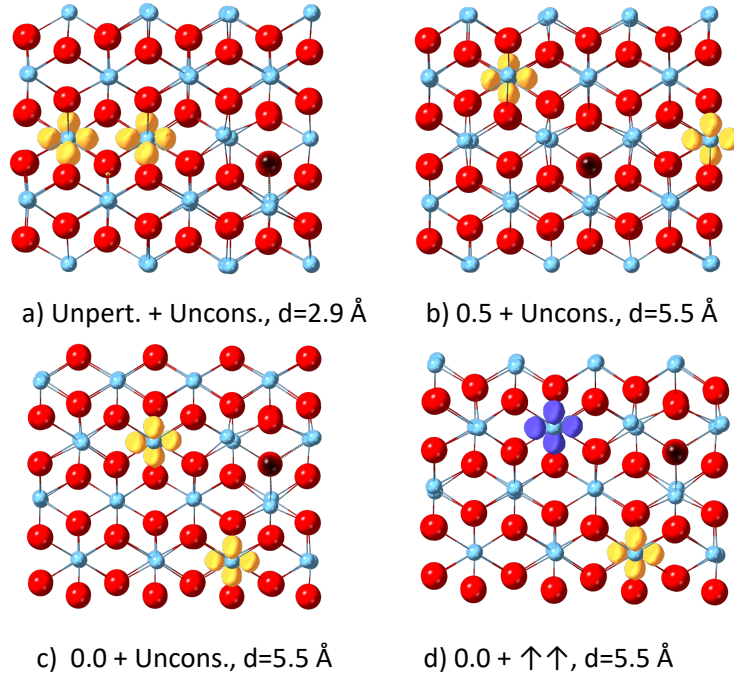
2. Neutral oxygen vacancy in TiO_2

A similar situation occurs for the neutral oxygen vacancy in rutile TiO_2 . Upon vacancy formation, the two electrons donated by the vacancy can localise on different Ti ions, reducing them to Ti(III) . To test whether our approach could find the different polaronic configurations and investigate the influence of enforcing magnetisation constraints, we applied the method with different setups: i) without any magnetisation constraints, ii) setting the number of unpaired electrons to 2 (spin-aligned, $\uparrow\uparrow$) and also to 0 (anti-aligned, $\uparrow\downarrow$). As shown in Figure 22, $\uparrow\uparrow$ and $\uparrow\downarrow$ show negligible energy differences, in agreement with previous studies^{40,45}.

The most favourable configurations correspond to the Ti(III) ions located in the second or third coordination shell (Fig. 23), as reported by a previous HSE06 study⁴⁴. This however differs from other computational studies which employed the Hubbard correction and found both Ti(III) to lie in the first coordination shell (N(1), N(1))⁴⁰. To verify our method was not missing this state, we performed additional geometry optimisations initialising the magnetic moments of two of the vacancy nearest neighbours. These configurations relaxed to a state with both Ti(III) ions located further away from the vacancy (first and second coordination shell), demonstrating that the N(1), N(1) state is not locally stable with this functional and supercell size, as previously reported by another HSE06 study⁴⁴. Finally, we note that all lowest energy states are successfully identified by the unconstrained run in this case, with comparable performance in the case of the spin-aligned ($\uparrow\uparrow$) constraint.



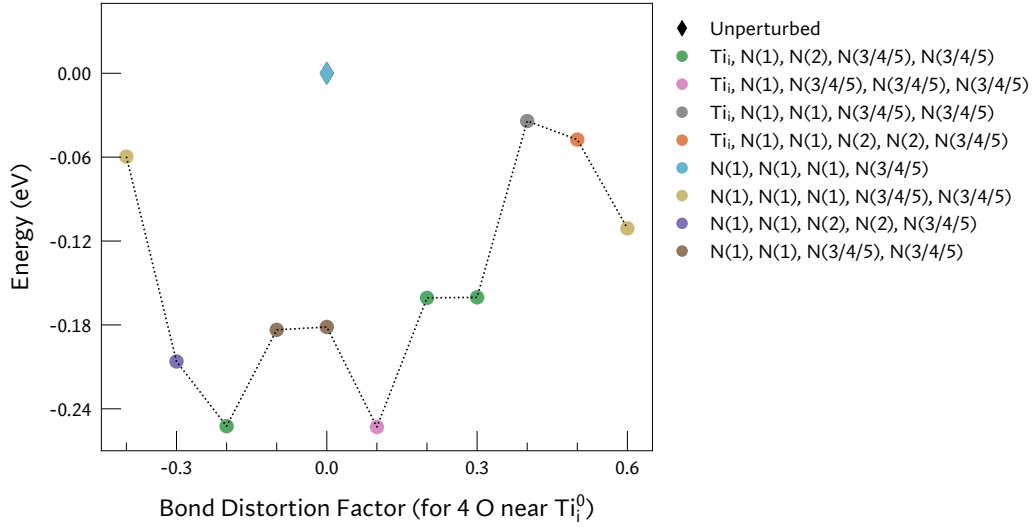
Supplementary Figure 22: Relative energy (in eV) of the different configurations identified for the neutral oxygen vacancy in rutile TiO_2 . The method was applied by either constraining the difference of the number of electrons in the up and down spin component to 2 (solid line), to 0 (dashed line) and without constraints (dotted line). The different colours indicate the distance between the two Ti(III) ions while the marker shapes correspond to where the Ti(III) are located relative to the vacancy: first coordination shell (N(1), i.e. nearest neighbour), second coordination shell (N(2), i.e. next nearest neighbour) and so on. As shown, spin-aligned and anti-aligned states have negligible energy differences.



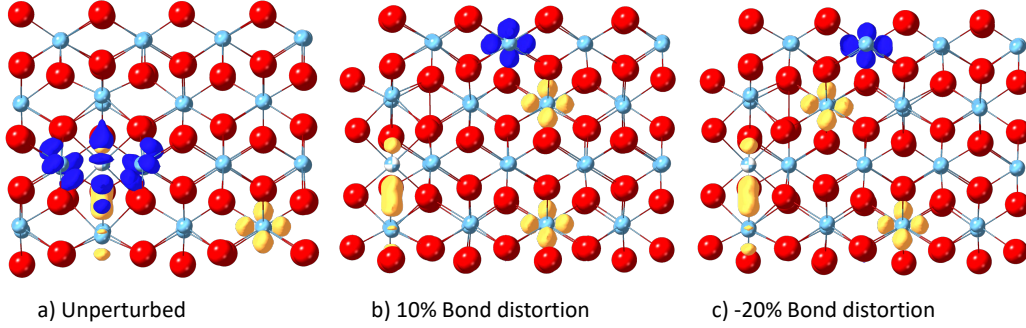
Supplementary Figure 23: Lowest energy configurations identified for V_O^0 in anatase TiO_2 . The labels indicate the bond distortion leading to that state, magnetisation constraints and distance between Ti(III) sites. The spin density associated to the states introduced by V_O^0 is shown, with different colours to differentiate spin channels. In all cases, the isosurfaces correspond to 25% of the maximum charge density. V_O in black, Ti in blue and O in red.

3. Neutral titanium interstitial in TiO_2

As with the oxygen vacancy, the titanium interstitial can lead to different states depending on where the donated electrons localise. The identified configurations are shown in Figure 24 and 25. Performing a geometry optimisation from the high-symmetry structure results in most of the charge localised on the interstitial and its Ti neighbours (1 electron shared between Ti_i and a neighbouring Ti, 2 electrons localised on two NN and 1 electron localised further away, Fig. 25 (a)). This state is similar to the configuration reported by previous hybrid^{42,44} and GGA+U studies⁵⁷. In contrast, our approach finds lower energy states where three of the donated electrons localise on Ti ions further away from the defect while the remaining electron is shared between the interstitial and a neighbouring Ti.



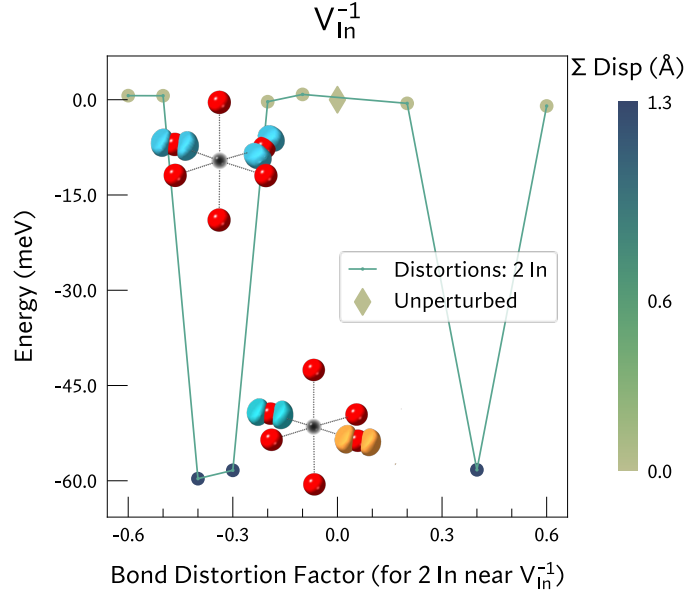
Supplementary Figure 24: Relative energy (in eV) of the different configurations identified for the neutral titanium interstitial in rutile TiO_2 . Standard relaxation from the high symmetry structure shown with a diamond (Unperturbed). The method was applied without magnetisation constraints. The different colours indicate the position of the Ti ions where the additional charge is localised, relative to Ti_i : first coordination shell (N(1)), second coordination shell (N(2)) and so on. We find negligible energy difference between $\uparrow\downarrow$ and $\uparrow\uparrow$ (2 unpaired electrons) arrangements. We note that certain configurations with similar localisation patterns differ in their energy due to small structural differences and/or the degree of charge localisation.



Supplementary Figure 25: Low energy structures identified for the neutral titanium interstitial in rutile TiO_2 . The spin density associated to the Ti_i states is shown, with different colors indicating different spin channels. In all cases, the isosurfaces correspond to 30% of the maximum charge density.

4. Singly negative indium vacancy in In_2O_3

Finally, we also considered the singly negative charged indium vacancy in In_2O_3 , where the method was applied without magnetisation constraints. As shown in Figure 26, a standard relaxation from the ideal structure results in a spin-aligned state ($\uparrow\uparrow$), with two holes localised on two of the vacancy neighbours. In contrast, we find an anti-aligned state ($\uparrow\downarrow$) to be more stable (by 60 meV). Compared to spin-aligned, in the anti-aligned case the two oxygens with a hole localised are displaced further away from each other (by 0.1 Å). While for this simple PES relaxing with magnetisation constraints would also identify the $\uparrow\downarrow$ state, these results demonstrate that applying the method without magnetisation constraints can successfully identify both configurations. In general, these results exemplify the ability of the method to identify different polaronic states, even for defects with a complex PES with several low energy minima.



Supplementary Figure 26: Relative energy (in meV) and structures of the different configurations identified for the singly negative indium vacancy in In_2O_3 . Standard relaxation from the high symmetry structure shown with a diamond (Unperturbed). The colourbar represents the structural difference between each structure and the ‘Unperturbed’ one, measured as the sum of the atomic displacements between structures. V_{In} in black and O in red. The spin density of different channels is shown with different colours.

F. Impact on defect properties

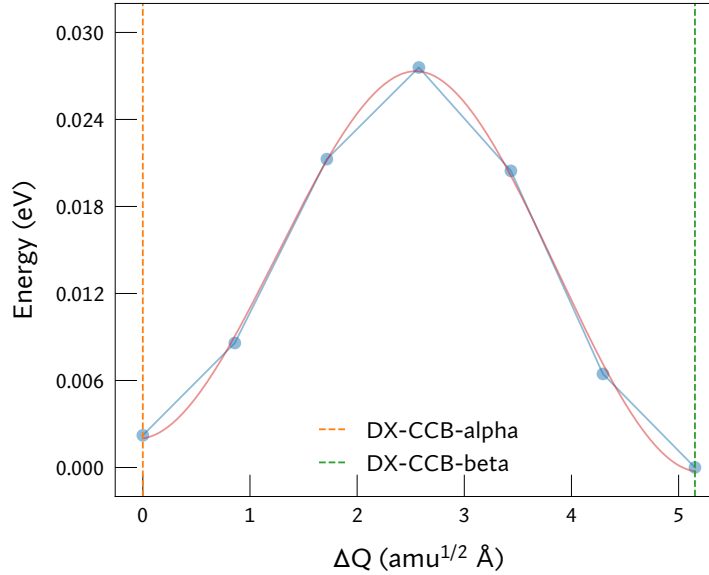
Supplementary Table IV: Concentrations (in cm^{-3}) of the stable charge states for the ground and metastable structures of V_{Sb} in Sb_2S_3 , for typical growth conditions ($T = 550$ K) and the Fermi level 0.97 eV above the VBM, as determined experimentally⁵⁸. Dashes (‘-’) signify unstable charge states.

$V_{\text{Sb},1}$			$V_{\text{Sb},2}$		
Charge	Ground state	Metastable	Charge	Ground state	Metastable
+2	2.1×10^{-2}	5.7×10^{-17}	+2	4.4×10^{-3}	5.0×10^{-22}
+1	4.2×10^2	-	+1	6.3×10^6	5.9×10^{-15}
0	-	-	0	-	1.8×10^{-7}
-1	8.4×10^8	2.3×10^3	-1	-	3.8×10^{-1}
-3	1.1×10^{14}	1.1×10^{14}	-3	1.1×10^{12}	1.1×10^{12}

Supplementary Table V: Positions of thermodynamic charge transition levels (in eV above the VBM) for the ground and metastable structures of V_{Sb} in Sb_2S_3 . Dashes (‘-’) signify thermodynamically-unstable transition levels.

$V_{\text{Sb},1}$			$V_{\text{Sb},2}$		
Level	Ground state	Metastable	Level	Ground state	Metastable
(2/1)	0.50	-	(2/1)	-0.03	-0.14
(2/-1)	-	0.26	(1/0)	-	0.15
(1/-1)	0.63	-	(0/-1)	-	0.28
-	-	-	(1/-3)	0.83	-
(-1/-3)	0.69	0.39	(-1/-3)	-	0.29

G. Identifying metastable structures



Supplementary Figure 27: Energy barrier (in eV) between $\text{DX} - \text{CCB} - \alpha$ and $\text{DX} - \text{CCB} - \beta$ for $\text{S}_{\text{As}}^{-1}$ in GaAs, calculated with the nudged elastic band method (5 images).

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